Thermal Shutdown System for Power MOSFET

Thermal Dynamics and Protection

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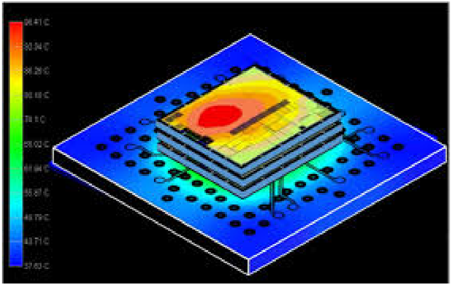
The Problem: Chip Heating

Why do chips heat up?

- Switching losses in CMOS gates
- Leakage current in scaled transistors
- High power density
- Localized thermal hot spots

What happens at high temperature?

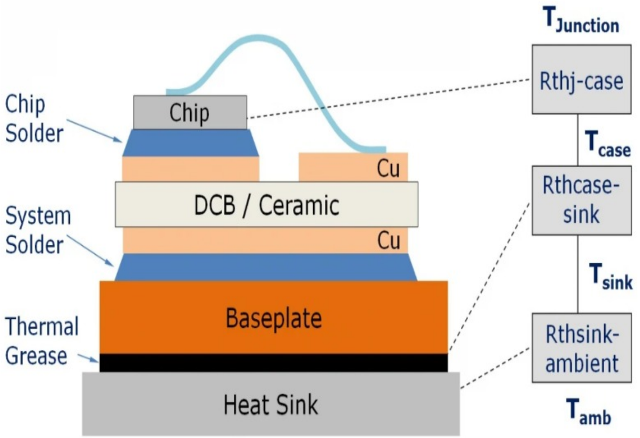
- Thermal runaway
- Reduced reliability
- Permanent silicon damage



Thermal hot-spot distribution in a 3D-IC package.

Source: SemiEngineering (Reela Samuel).

Physical Structure and Thermal Paths



Cross section of a typical module showing the different thermal resistance values

Source: Jeremy Howes, “Temperature Limits for Power Modules Part-1: Maximum Junction Temperature,” EE Power, January 01, 2016.

Device and Thermal Parameters(Infineon Data Sheet)

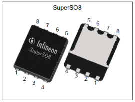
BSC009NE2LS5

MOSFET

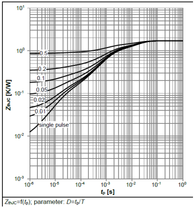
OptiMOS™ 5 Power-MOSFET, 25 V

Features

- Optimized for high performance buck converters
- Very low on-resistance $R_{DS(on)}$ @ V_{GS}=4.5 V
- 100% avalanche tested
- Superior thermal resistance
- N-channel
- Qualified according to JEDEC[1] for target applications
- Pb-free lead plating; RoHS compliant
- Halogen-free according to IEC61249-2-21



NMOS power MOSFET selected for the thermal shutdown prototype.



Transient thermal impedance $Z_{th}(t)$ curve showing the dynamic thermal response of the MOSFET from junction to case across pulse durations.

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2 Thermal characteristics

Table 3 Thermal characteristics

Parameter	Symbol	Values			Unit	Note / Test Condition
		Min.	Typ.	Max.		
Thermal resistance, junction - case, bottom	R_{thJC}	-	-	1.7	K/W	-
Thermal resistance, junction - case, top	R_{thJC}	-	-	20	K/W	-
Device on PCB, 6 cm^2 cooling area[2]	R_{thJA}	-	-	50	K/W	-

Extracted Foster RC network values (R_{th}, C_{th}) modelling the junction-to-case thermal path of the MOSFET.

Thermal RC Analogy

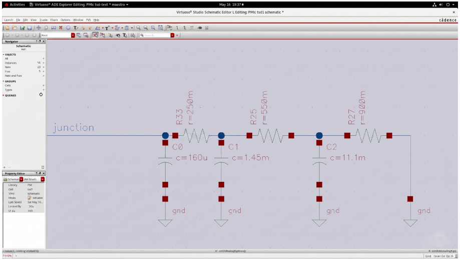
Thermal System

- Temperature $\leftrightarrow$ Voltage
- Power $\leftrightarrow$ Current
- Thermal Resistance $\leftrightarrow$ Resistance
- Heat Capacity $\leftrightarrow$ Capacitance

$$\tau = R_{th} C_{th}$$

Heat spreads slowly like a capacitor

charging.



Thermal RC ladder modelling heat flow as an electrical circuit,where R_{th} and C_{th} represent thermal resistance and heat capacity respectively.

Stable Voltage Reference: The Bandgap (BGR)

The Challenge

- As the MOSFET operates, chip temperature fluctuates significantly.
- A shutdown system requires a **stable reference** threshold to compare against the sensor.
- Standard references shift with temperature, leading to inaccurate trip points.

Thermal Compensation

BGR combines two opposing thermal behaviors to cancel out variation:

- **CTAT:** V_{BE} drops as temperature rises.
- **PTAT:** ΔV_{BE} increases as temperature rises.

The BGR Solution

- Generates a voltage ($\approx 1.25\,\text{V}$) with a **near-zero temperature coefficient**.
- Acts as the "Fixed Ruler" that does not expand or shrink with heat.

Summation Formula:

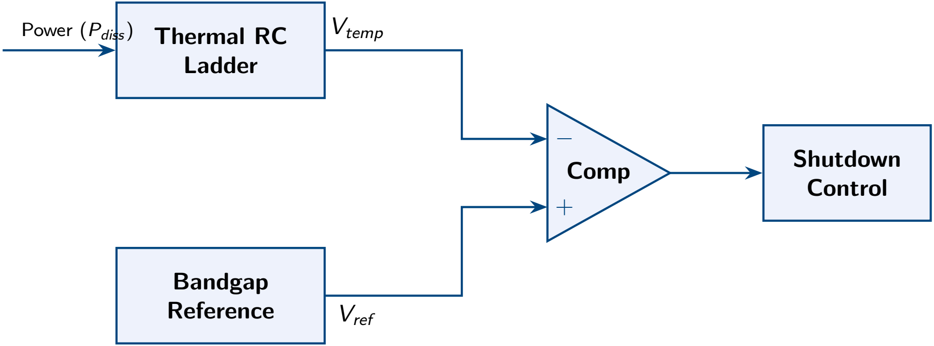
$$V_{BGR} = \underbrace{V_{BE}}_{\text{CTAT}} + \underbrace{K \cdot V_T}_{\text{PTAT}}$$

$$\text{Result: } \frac{\partial V_{BGR}}{\partial T} \approx 0$$

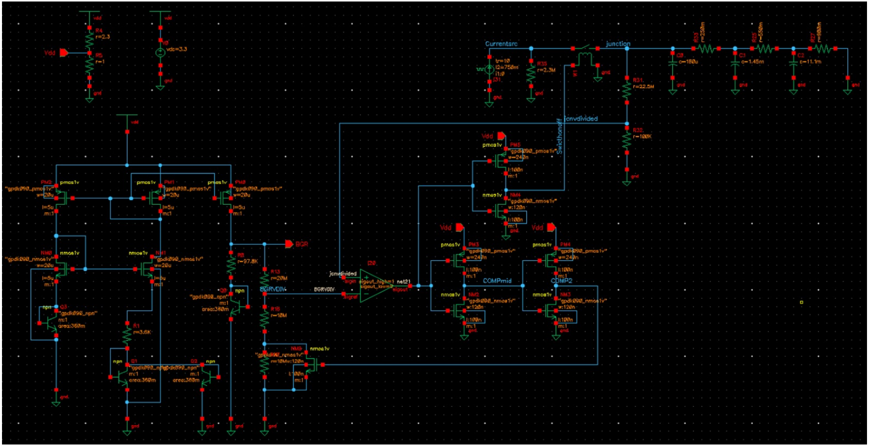
Bandgap Reference (BGR) Characteristics

Temperature compensation plot demonstrating the summation of CTAT and PTAT voltage components to achieve a stable, constant Bandgap Reference (BGR) output across a wide thermal range.

System Control Block Diagram



Complete System Schematic



Complete circuit schematic of the thermal shutdown system, showing the thermal RC ladder, bandgap reference, and comparator stages.

Comparator and Hysteresis

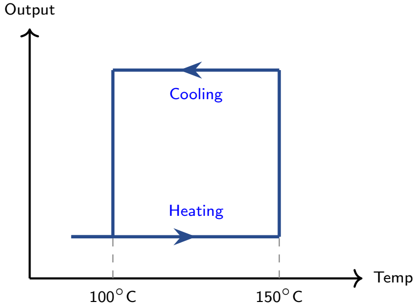
Without Hysteresis

- Near threshold: $ON \leftrightarrow OFF$
- Rapid switching causes **chattering** and electrical noise.

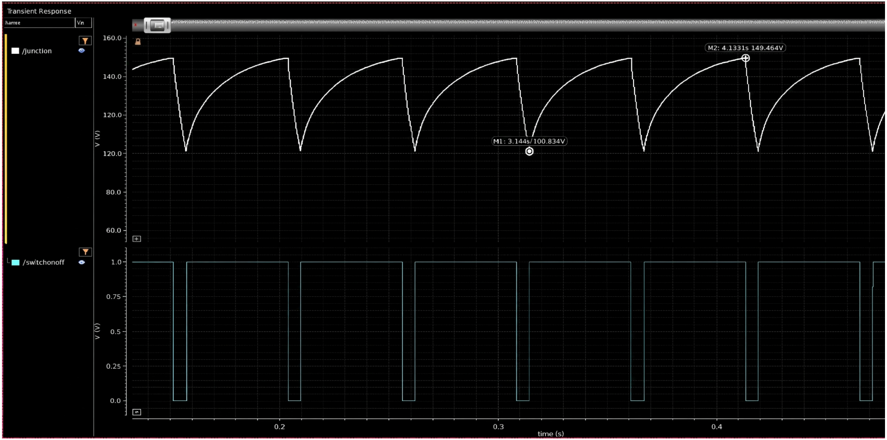
With Hysteresis (Dual Threshold)

- **Shutdown:** Activates at 150°C
- **Restart:** Re-enables at 100°C

Ensures stable operation within the 100°C–150°C window.



Transient Response: Thermal Shutdown Cycling



A transient simulation with a one-to-one temperature-to-voltage mapping (100°C = 100V). The plot shows junction temperature (top) cycling between 100°C and 150°C, as the control signal `/Switchonoff` (bottom) toggles between ON and OFF, confirming correct hysteresis.

Conclusion

- Thermal behavior modeled using RC analogy
- BGR provides stable reference
- Comparator + hysteresis prevents chattering
- Verified in Cadence Virtuoso

Thank You